

Question:

Tom's age is \$T\$ years, which is also the sum of the ages of his three children. His age \$N\$ years ago was twice the sum of their ages then. What is \$T/N\$?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Reasoning + Answer:

<think>

1. Understand the given information :
- Tom 's current age is (T) years .

- The sum of the ages of his three children is also (T) .

- (N) years ago , Tom 's age was twice the sum of the ages of his children at that time .
2. Define the ages of the children :
- Let the ages of the three children be (a) , (b) , and (c) .

- Therefore , $(a + b + c = T)$.
3. Determine Tom 's age and children 's ages (N) years ago :
- (N) years ago , Tom 's age was $(T - N)$.

- The ages of the children (N) years ago were $(a - N)$, $(b - N)$, and $(c - N)$.

- The sum of the children 's ages (N) years ago was $((a - N) + (b - N) + (c - N) = (a + b + c) - 3 N = T - 3 N)$.
4. Set up the equation based on the given condition :
- According to the problem , (N) years ago , Tom 's age was twice the sum of the children 's ages at that time .

- Therefore , $(T - N = 2 (T - 3 N))$.
5. Solve the equation for (T) in terms of (N) :
- $(T - N = 2 T - 6 N)$

- $(T - 2 T = - 6 N + N)$

- $(- T = - 5 N)$

- $(T = 5 N)$
6. Find the ratio $(\frac { T } { N })$:
- $(\frac { T } { N } = \frac { 5 N } { N } = 5)$
- </think>
- <answer>
- 5
- </answer>